SSE/POWER BCL

MUMBAI DIVISION WESTERN RAILWAY

Subject: **AI Smart Energy Saving Device Report**

Respected Sir,

I, DEVENDRA UMESH CHAVAN, POST GRADUATE in Data Science. Working on an Artificial Intelligence Smart Energy Saving Device for Indian Railways.

I have Demonstrated Smart Device on a single Fan at Mumbai Central Railway Station (Local) on Platform No 3 for 1 month. Device Automatically Switch OFF the Fan if not in use.

Installed device on 2 December 2023 to 2 jan 2024

Fan Power ON (5 am to 1.30 am)

**POWER SAVING** Report for **1 Month** (30 watt fan)

**Total Power Consumed**

| | Ordinary Fan | Fan with (SMART DEVICE) |
|---|---|---|
| WATT | **19,050** | **3,880** |
| UNITS | **19** | **3.9** |

**Total Energy SAVED**

| | |
|---|---|
| WATT | **15,170** |
| UNITS | **15.2** |

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Total Electricity **Saved 15.2 units** in 1 month **(79.6 %)** as compare to other Fan total electricity consumption without Smart Device Installed

**Power Saving Cost Calculation**

| Unit Rate | Units Saved | Monthly Saving | Yearly Saving |
|---|---|---|---|
| 8.1 ₹ | 15.2 | 123 ₹ | 1477 ₹ |

**Return on Investment (ROI)**

| Product Cost (Approx) | 2000 ₹ | 2500 ₹ | 3000 ₹ |
|---|---|---|---|
| Year | 1.4 | 1.8 | 2.1 |

Thank You

Yours Sincerely

Devendra Chavan

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